

No. of Functional Features Included in the Solution

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Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Authentication	Allows users and administrators to securely log in to the system.	Authentication	Done	High
2	Dataset Upload	Enables administrators to upload HDI datasets in CSV format.	Dataset Management	Done	High
3	Data Preprocessing	Cleans missing values, normalizes data, and prepares features for prediction.	Data Processing	Done	High
4	HDI Prediction	Predicts the Human Development Index category using a trained Machine Learning model.	Prediction Engine	Done	High
5	Country Comparison	Compares HDI values and development indicators between multiple countries.	Analysis Module	Done	Medium
6	Interactive Dashboard	Displays prediction results through charts, graphs, and summary statistics.	Dashboard	Done	High

7	Report Generation	Generates downloadable prediction reports in PDF and CSV formats.	Reporting Module	Done	Medium
8	Prediction History	Stores previous predictions for future reference and analysis.	Database Module	Done	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Authentication, Dataset Upload, Data Preprocessing, HDI Prediction, Dashboard)
Additional / Nice-to-Have Features	3 (Country Comparison, Report Generation, Prediction History)
Features Tested & Verified	8 / 8 (100%)

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Login Page, HDI Input Form, Interactive Dashboard
2	Backend / Logic	3	Data Preprocessing, Machine Learning Prediction, Business Logic
3	Database / Storage	2	User Information, Prediction History
4	API / Integration	2	Flask REST API, Dataset Import (CSV)
5	Security / Authentication	2	User Login, Password Validation

Functional Features Summary

The **Human Development Index (HDI) Prediction System** includes all the essential functional features required for accurate and efficient HDI prediction. The implemented features cover secure authentication, dataset management, machine learning-based prediction, interactive dashboards, report generation, and historical data storage. These functionalities provide users with a complete, reliable, and user-friendly platform for analyzing human development indicators and supporting data-driven policy decisions.

Figure 10.1: Functional Features Included in the HDI Prediction System



